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```
import pandas as pd

data = {
    'Weight':[79,69,73,95,82,55,69,71,64,69],
    'Height':[1.80,1.68,1.82,1.70,1.87,1.55,1.50,1.78,1.67,1.64],
    'Age':[35,39,25,60,27,18,89,42,16,52],
    'Gender':['Male','Male','Male','Male','Male','Female','Female','Female',
}

df = pd.DataFrame(data)
df
```

	Weight	Height	Age	Gender
0	79	1.80	35	Male
1	69	1.68	39	Male
2	73	1.82	25	Male
3	95	1.70	60	Male
4	82	1.87	27	Male
5	55	1.55	18	Female
6	69	1.50	89	Female
7	71	1.78	42	Female
8	64	1.67	16	Female
9	69	1.64	52	Female

Next steps: Generate code with df | New interactive sheet

```
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_scor
```

```
df['Gender'] = df['Gender'].map({'Male':1, 'Female':0})

df
```

	Weight	Height	Age	Gender
0	79	1.80	35	1
1	69	1.68	39	1
2	73	1.82	25	1
3	95	1.70	60	1
4	82	1.87	27	1
5	55	1.55	18	0
6	69	1.50	89	0
7	71	1.78	42	0
8	64	1.67	16	0
9	69	1.64	52	0

Next steps: Generate code with df | New interactive sheet

```
X = df[['Height','Age','Gender']]
y = df['Weight']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
model = LinearRegression()

model.fit(X_train, y_train)
```

```
▾ LinearRegression
LinearRegression()
```

```
y_pred = model.predict(X_test)

print(y_pred)
```

```
[56.85651025 79.10576115]
```

```
print("MAE :", mean_absolute_error(y_test, y_pred))
print("MSE :", mean_squared_error(y_test, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
print("R2 Score:", r2_score(y_test, y_pred))
```

```
MAE : 8.624625451606978
MSE : 76.57792714844265
RMSE: 8.750881506936468
R2 Score: -11.252468343750825
```

```
new_person = [[1.75, 24, 1]]

prediction = model.predict(new_person)

print("Predicted Weight:", prediction[0])
```

```
Predicted Weight: 76.64481356828415
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: Use
  warnings.warn(
```

```
print("Intercept:", model.intercept_)
print("Coefficients:", model.coef_)
```

```
Intercept: -3.0552172392595196
Coefficients: [32.8353363   0.31729474 14.62311848]
```